B2.1 Membranes and membrane transport

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

7 MARKS

1 DEFINE amphipathic. [B2.1.1] [1]

2 STATE two gases that cross membranes by simple diffusion. [B2.1.3] [1]

3 DISTINGUISH between integral and peripheral proteins. [B2.1.4] [2]

4 OUTLINE the role of aquaporins. [B2.1.5] [1]

5 STATE two roles of glycoproteins and glycolipids. [B2.1.9] [2]

Section B: Understanding

15 MARKS

6 EXPLAIN why phospholipids form bilayers in water. [B2.1.1] [3]

7 EXPLAIN why a phospholipid bilayer is a barrier to ions. [B2.1.2] [2]

8 EXPLAIN osmosis, using the terms membrane and solute. [B2.1.5] [3]

9 **COMPARE AND CONTRAST** facilitated diffusion and active transport. [B2.1.6, B2.1.7] [4]

FACILITATED DIFFUSION	ACTIVE TRANSPORT

10 **EXPLAIN** how a membrane can be selectively permeable. [B2.1.8] [3]

Section C: Working with data

12 MARKS

All data in this section are invented for practice. The patterns are realistic.

11 [B2.1.3, B2.1.6]

The rate at which two substances entered cells was measured at a range of outside concentrations.

OUTSIDE CONCENTRATION / MMOL DM^{-3}	1	2	4	8	16
SUBSTANCE X: RATE / ARBITRARY UNITS	2	4	8	16	32
SUBSTANCE Y: RATE / ARBITRARY UNITS	5	9	14	16	16

a **DESCRIBE** the trend for each substance. [2]

b **DEDUCE**, with a reason, which substance crosses by simple diffusion. [2]

c **EXPLAIN** why the rate for the other substance stops rising. [2]

12 [B2.1.7, B2.1.8]

Slices of carrot were kept in a solution of potassium ions. The uptake of potassium was measured with and without a substance that stops cell respiration.

CONDITION	POTASSIUM ION UPTAKE / MMOL G^{-1} H^{-1}
NO INHIBITOR	12.0
WITH INHIBITOR	1.5

a CALCULATE the percentage decrease in uptake. [1]

b EXPLAIN what these results suggest about how potassium ions enter the cells. [3]

c SUGGEST why some uptake continued with the inhibitor. [2]

Section D: Longer answer

6 MARKS

13 DRAW a labelled diagram of the fluid mosaic model of a membrane. [B2.1.10] [6]